DATA SCIENCE & MACHINE LEARNING

Guest Intelligence System

Recommender System for Personalized Hotel Experiences

Bootcamp Final Project

PRESENTATION AGENDA

Agenda

01 **What is GIS?**
Understanding the Guest Intelligence System concept and its three pillars

02 **Problem Definition**
The critical gap in current hospitality systems

03 **Project Motivation**
Why this project matters in the hospitality industry

04 **The 5 Problems**
Critical hospitality challenges GIS addresses

05 **System Architecture**
Database design and technical implementation

06 **Recommender Models**
Content-based, collaborative, and hybrid approaches

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User interface and practical application

01

Understanding the System

What is Guest Intelligence System and how it transforms hospitality through data-driven personalization

SYSTEM OVERVIEW

What is Guest Intelligence System?

The **Guest Intelligence System (GIS)** is an intelligent platform that learns each guest's profile — their preferences, needs, constraints, and history — and uses this knowledge to personalize experiences in real-time, prevent problems before they happen, and ensure every trip is better than the previous one.

Behavioral KYC

Know Your Customer approach adapted for hospitality

- Evaluates and knows the guest
- Detects anomalies and assesses risk
- Protects the company from fraud

In Development

Recommender System

Personalizes the experience like Netflix or Spotify

- Suggests activities based on profile
- Recommends upgrades and services
- Adapts to guest preferences

Implemented

Behavioral Profile

Accumulates data across multiple stays

- Learns from each stay
- System improves over time
- Creates persistent guest memory

Implemented

Current Implementation: The GIS currently features a recommender system that suggests activities based on guest profiles and similar guest behaviors, anticipating special occasions and needs without the guest having to mention them — just by learning from previous stays.

PROBLEM DEFINITION

The Problem We're Solving

> "Current hospitality and cruise systems don't store or use accumulated knowledge about each guest — forcing customers to always repeat the same information, and preventing companies from anticipating problems and personalizing experiences."

Current State

1. **No Guest Memory**
   Each booking is treated as a first-time guest
2. **Information Repetition**
   Guests must re-enter preferences every time
3. **Reactive Approach**
   Problems solved after they occur
4. **Generic Experiences**
   One-size-fits-all service model

With GIS

1. **Persistent Profiles**
   Every stay enriches guest knowledge
2. **Automatic Recognition**
   Preferences applied without asking
3. **Proactive Prevention**
   Issues anticipated and prevented
4. **Hyper-Personalized**
   Tailored to each individual guest

3,500+	7	81
Guest Profiles	Interconnected Tables	Activities Catalog

PROJECT MOTIVATION

Why This Project?

The hospitality industry has a critical gap that major platforms ignore. While **Booking, Airbnb, and Expedia** solve the problem of **choosing destinations** (BEFORE the trip), they completely miss the **DURING and AFTER** trip experience — and this is where the real opportunity lies.

BEFORE Trip

SATURATED MARKET

- Airbnb
- Booking.com
- Expedia
- TripAdvisor

Focus: Discovery & Booking

DURING Trip

THE GAP

- Real-time Personalization
- Problem Prevention
- Anticipatory Service
- Guest Recognition

Focus: Experience Optimization

AFTER Trip

UNDERSERVED

- Profile Enrichment
- Learning & Improvement
- Next Trip Preparation
- Loyalty Building

Focus: Memory & Learning

The Opportunity: While existing platforms focus on getting guests TO the hotel, GIS focuses on what happens DURING and AFTER the stay — creating a complete, personalized experience that builds loyalty and increases revenue.

02

The Five Problems

Critical hospitality challenges that the Guest Intelligence System addresses through data science and machine learning

HOSPITALITY CHALLENGES

The Five Problems in Hospitality

1 Special Requests Not Saved

IMPLEMENTED

Problem: Guests must re-enter allergies and special requests for every reservation. The system doesn't retain information between trips.

GIS Solution: Persistent guest profile that automatically stores all preferences, allergies, and special requests — available for all future stays without repetition.

2 Complaints Resolved After Trip

IN DEVELOPMENT

Problem: Complaining guests are advised to call customer service after returning home. But the bad trip is already over — the experience cannot be undone.

GIS Solution: Real-time dissatisfaction detection system — when a complaint pattern is detected, immediately alerts staff to resolve DURING the trip.

3 Guest Financial Profile Ignored

IN DEVELOPMENT

Problem: If it's the trip of a lifetime, the company doesn't know — and does nothing special. A small gesture would cost little but create a memory forever.

GIS Solution: Automatically identify high emotional value guests (not just financial) and trigger personalized gestures — upgrades, special welcomes, extra service.

4 No Adaptability for Reduced Mobility

IMPLEMENTED

Problem: The system doesn't automatically block rooms or excursions inaccessible to guests with mobility difficulties — creating embarrassing and dangerous situations.

GIS Solution: Guest profile includes accessibility needs — the recommender automatically filters and only suggests compatible options. Adapted excursions are prioritized.

5 Generic Activity Suggestions

IMPLEMENTED

Problem: Activity suggestions are generic — they don't consider guests' real needs, limitations, or tastes. A guest with difficulties might want to experience something "radical" in an adapted way.

GIS Solution: Recommender that goes beyond obvious preferences — finds creative ways to provide special experiences within guest limitations using collaborative and content-based filtering.

SYSTEM OPERATION

System Phases: During & After Stay

The GIS operates primarily in **Phase 2 (During Stay)** — the main focus where the system observes real-time behavior and takes action. Phase 3 ensures continuous improvement for future stays.

Phase 2
DURING Stay

This is the most important phase where the project has maximum impact.

Guest goes to spa today
→ Auto-book spa for tomorrow

Guest makes complaint
→ Alert staff to resolve NOW

Guest with reduced mobility
→ Filter next excursion options

"Trip of a lifetime" guest
→ Trigger special gesture

Real-time Action & Observation

Phase 3
AFTER Stay

The guest profile is updated with everything that happened during this trip.

Profile Enrichment
All preferences and behaviors stored

Machine Learning Update
Models retrained with new data

Pattern Recognition
New insights about guest preferences

Next Trip Preparation
System ready for future bookings

Continuous Learning & Improvement

The Magic: On the next reservation, the system already knows more — and the experience feels "magical" to the guest because preferences are applied without asking.

03

Technical Implementation

Database architecture, machine learning models, and the recommender system development

DATABASE DESIGN

Database Architecture

The GIS database was built using **Python connected to SQLite**, with 7 interconnected tables storing comprehensive guest information.

Libraries used: Faker, Pandas, SQLite3, Unicodedata, NumPy, Random.

#	Table Name	Description	Records	Raw Columns	After FE
1	GUEST_PROFILE	Guest identity and preferences	3,500	31	27
2	BOOKING_HISTORY	Reservation and payment data	5,000	30	40
3	STAY_BEHAVIOR	Operational behavior during stay	3,021	25	39
4	REVIEWS	Post-stay guest reviews	2,095	37	30
5	ACTIVITY_CATALOG	Master catalog of available activities	81	24	22
6	COMPLAINTS_LOG	Detailed complaint records	1,698	21	21
7	ACTIVITY_PREFERENCES	Guest activity preferences & accessibility	3,021	32	41

7
Interconnected Tables

18,416
Total Records

SQLite
Database Engine

Python
Development

RECOMMENDER SYSTEM

Recommender System: Three Models

A **Recommender System** suggests things to a user based on their past tastes and behaviors — like Netflix or Spotify, but applied to the guest experience during and after the stay. **Objective:** Increase customer satisfaction and company revenue.

Model 1

Content-Based Filtering

Recommends based on the guest's own profile characteristics and preferences.

- Uses cosine similarity
- Maps 6 guest features
- Profile-activity matching

Metric: Cosine Similarity

Model 2

Collaborative Filtering

Recommends based on similar guest profiles and their behavior patterns.

- Finds similar guests
- SVD & Cornac BPR
- Best: WBPR algorithm

Metric: NDCG@10 = 0.8386

Model 3

Hybrid Model

Combines both approaches for optimal recommendations with balanced weights.

- Content + Collaborative
- CB=0.4 | CF=0.6
- Best of both worlds

Output: Hybrid Score

Output: List of personalized recommendations per guest with activity_id, category, and confidence scores — enabling the chatbot to suggest the most relevant activities.

MODEL 1: CONTENT-BASED

Content-Based Filtering

This model **recommends based on the guest's profile**, using cosine similarity to measure how compatible the guest profile is with each activity. We map 6 guest features to activity features and calculate similarity scores.

Guest Profile Features

1. preferred_price_range
2. is_family
3. is_solo
4. is_couple
5. kids_activities_needed
6. wellness_rating_avg

Activity Profile Features

1. price_range
2. suitable_for_families
3. suitable_for_solo
4. suitable_for_couples
5. suitable_for_kids
6. is_wellness

Recommendations for Guest G00004

ACT073 (Category 7)	0.9847
ACT038 (Category 1)	0.7590

Recommendations for Guest G00001

ACT037 (Category 1)	0.6201
ACT038 (Category 1)	0.6201

MODEL 2: COLLABORATIVE FILTERING

Collaborative Filtering Journey

This model **recommends based on similar guest profiles**, finding guests with similar behavior and recommending what they enjoyed most. We use SVD and Cornac BPR algorithms to find hidden patterns.

The Sparsity Problem

Original Dataset

Interactions: **7,329**

Unique Guests: **2,030**

Activities: **81**

Sparsity: **95.8%**

Solution: Feature Enrichment

Added 3 more features to create more interactions:

- spa_usage_flag → spa_visits
- excursion_usage_flag → excursion_visits
- bar_usage_flag → bar_visits

Enriched Dataset

Interactions: **66,010**

Sparsity: **61.4%**

Algorithm Performance

SVD (Initial) — Poor

Very low scores due to 95.8% sparsity

Cornac BPR — Good

NDCG@10 = 0.7842 with enriched data

WBPR (Best) — Best

NDCG@10 = 0.8386

Precision@10 = 0.5239 | Recall@10 = 0.8213

Other Algorithms — Poor

MF: 0.0758 | NMF: 0.0179

🏆 **Best Model:** WBPR (Weighted Bayesian Personalized Ranking) with NDCG@10 = 0.8386 — selected for the hybrid model

MODEL 3: HYBRID APPROACH

Hybrid Model

The **Hybrid Model** combines Content-Based and Collaborative Filtering to leverage the strengths of both approaches. It predicts recommendations based on guest preferences AND similar guest profiles.

Weight Configuration

Content-Based (CB) 0.4

Guest's own profile and preferences

Collaborative (CF) 0.6

Similar guests' behaviors and preferences

Hybrid Score = (0.4 × CB) + (0.6 × CF)

Hybrid Recommendations for G00001

ACT038 (Category 1)	CB: 1.0000	CF: 0.9174	0.9505
ACT076 (Category 1)	CB: 1.0000	CF: 0.9137	0.9482
ACT037 (Category 1)	CB: 1.0000	CF: 0.8877	0.9326
ACT067 (Category 6)	CB: 0.6284	CF: 0.6574	0.6458
ACT065 (Category 6)	CB: 0.6659	CF: 0.5771	0.6126

Personal Preferences

Content-Based ensures personal taste alignment

Social Proof

Collaborative adds discovery from similar guests

Optimal Balance

Hybrid combines best of both approaches

PROJECT SUMMARY

Conclusions & Future Work

Achievements

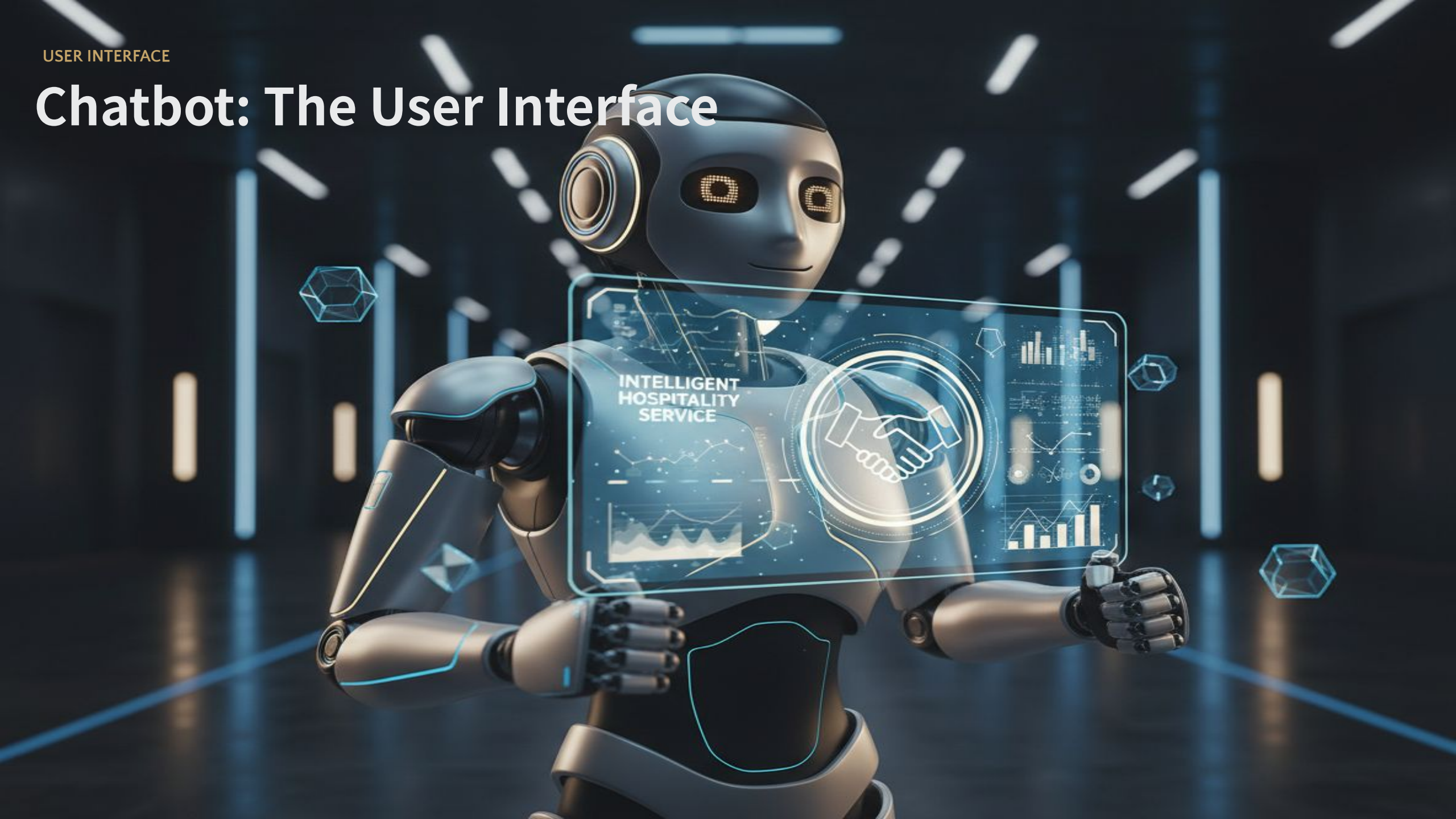
- Built comprehensive database with **7 interconnected tables** and 18,416+ records
- Developed **hybrid recommender system** achieving NDCG@10 = 0.8386
- Created **intelligent chatbot** interface for guest interaction
- Solved **3 of 5 identified problems** in hospitality

Future Work

- Complete **Behavioral KYC** for fraud detection and risk assessment
- Implement **real-time complaint detection** and resolution system
- Add **sentiment analysis** for emotional value detection
- Deploy **production system** with live guest data

The **Guest Intelligence System** demonstrates how Data Science and Machine Learning can transform hospitality by creating truly personalized experiences that learn and improve with every stay.

3/5	0.8386	18,416+
Problems Solved	NDCG@10 Score	Database Records

USER INTERFACE

Chatbot: The User Interface